

papa reo

The Creation of a Bilingual Māori-New Zealand English ASR

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te hiku media

Outline

- A history of Māori ASR models
- Why Bilingual
- Data Sources
- Conformer
- Training experiences
- Benchmarking and comparing models
- Wrap up

History of Māori capable ASR

Model	Who	Method	Trained on (hrs)	~ Num Params	Languages	Year	kWER *
DeepSpeech	Te Hiku Media & Mozilla	RNN + CTC	~ 400	47 m	Māori only	2019	~ 27 %
Whisper	Open AI	Enc/Dec Transformer	~ 680,000	1,550 m	97 *incl Māori*	2022	53 %
Conformer	Te Hiku Media & Nvidia	Conformer + RNNT	~ 5,000	120 m	Bilingual *Māori / English*	2023	10 %
MMS	Meta AI	Wav2Vec + CTC	~ 491,000	1,000 m	1000+ *incl Māori*	2023	46 %

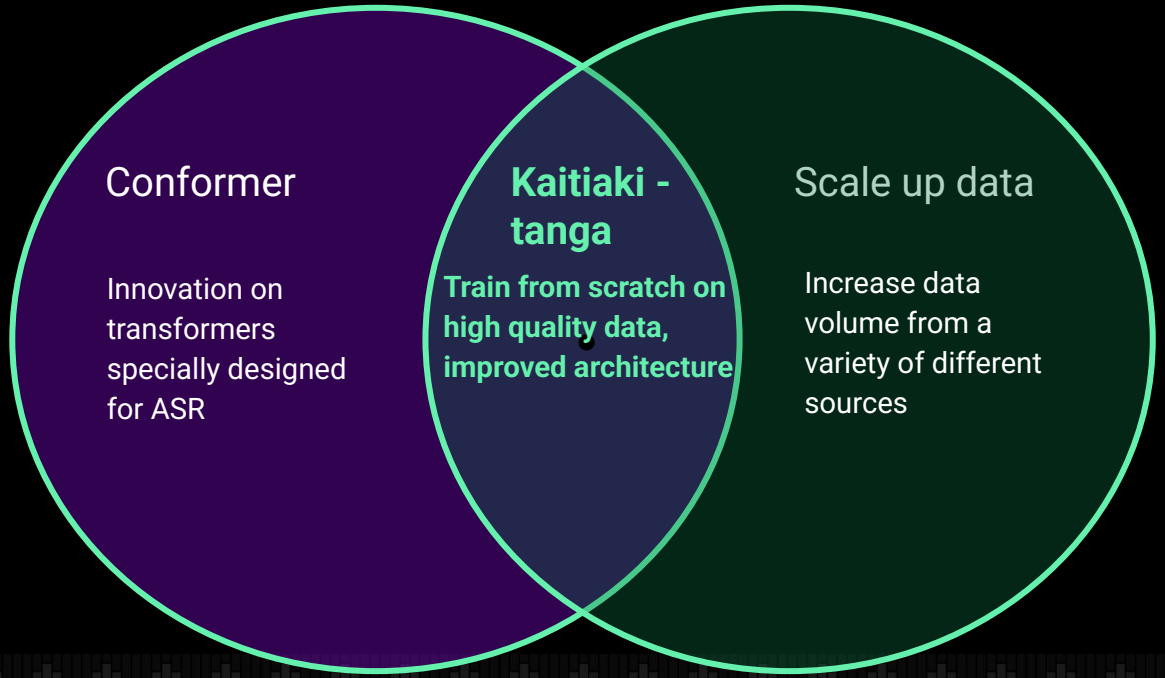
* Indicative word error rate on **contemporary Māori benchmark only**. See later slide for details.

Why Bilingual?



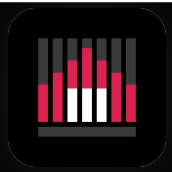
Our Approach

Train a model from scratch



Data

Expanding our train dataset



Whāre Kōrero

- Bilingual
- Local Iwi radio broadcast and communicatio ns hub.
- Automatic Transcription



Quakebox

- NZ English
- monologues about the 2010-2011 Canterbury earthquakes.
- Human Transcription



Kōrero Māori

- Te Reo Māori
- Community driven corpus
- Human Transcription



Kaituhi

- Bilingual
- Radio Transcripts
- Automatic transcription



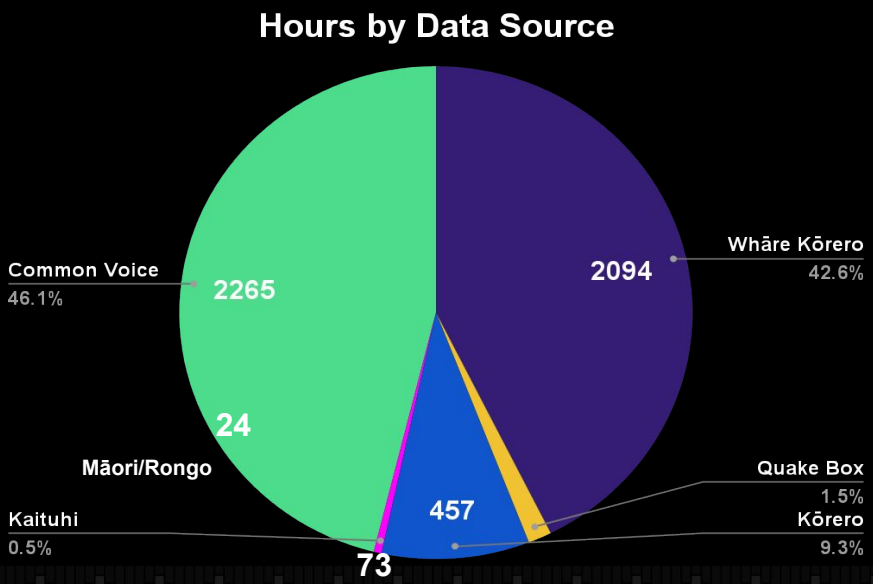
Common Voice

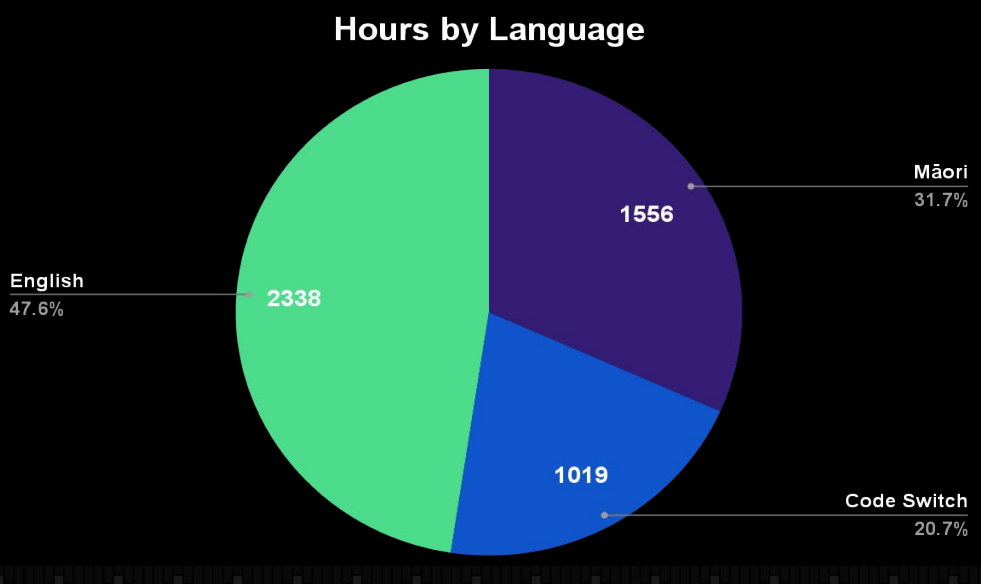
- English
- Crowd Sourced readings
- Human Transcription

Liam Walsh et al (2013). The UC QuakeBox Project: Creation of a community-focused research archive

Training Data

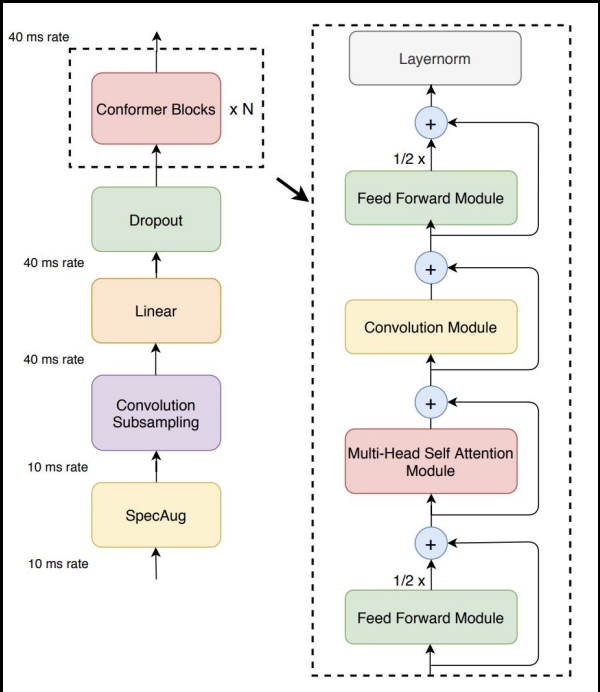
Expanding our train dataset





Conformer

Encoder

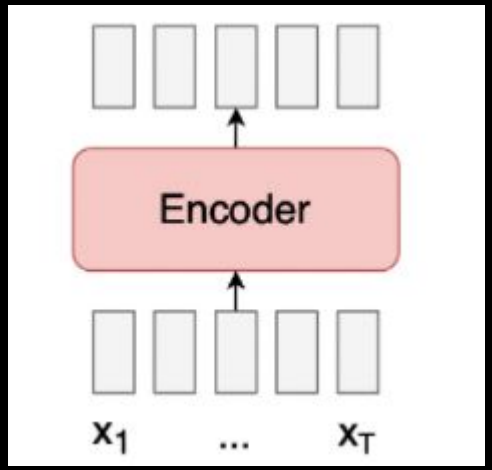


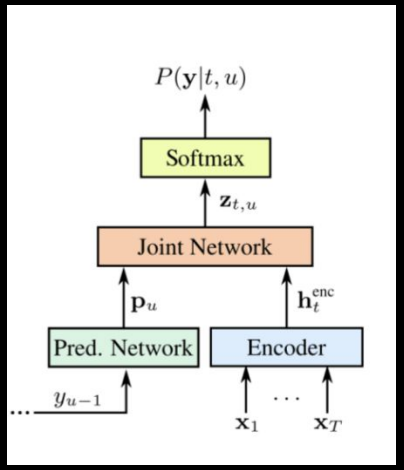
- Neural net for Automatic Speech Recognition (ASR)
- Builds upon Transformer architecture.
- Integration of Convolutional layers
- Captures both local and global dependencies.

Conformer: Convolution-augmented Transformer for Speech Recognition Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahui Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, Ruoming Pang (2020)

Conformer

CTC vs RNN-Transducers





- Predictions are conditionally independent
- "I have for apples" is as valid as "I have four apples"

- Encoder models acoustic features
- Predictor acts as language model
- Joint network combines Encoder and Predictor predictions to produce a label

Sequence Transduction with Recurrent Neural Networks.
Recognition Alex Graves (2012)

Training Conformer

Training experiences

- **Data Usage**

  Approximately 5,000 hours of comprehensive training data

- **Parameter and Size Metrics**

  Main Model: 120 Million parameters | Size: 450Mb
  Small Model: 12 Million parameters | Size: 50Mb

- **Training Duration**

  Approximately 6 days for complete model training

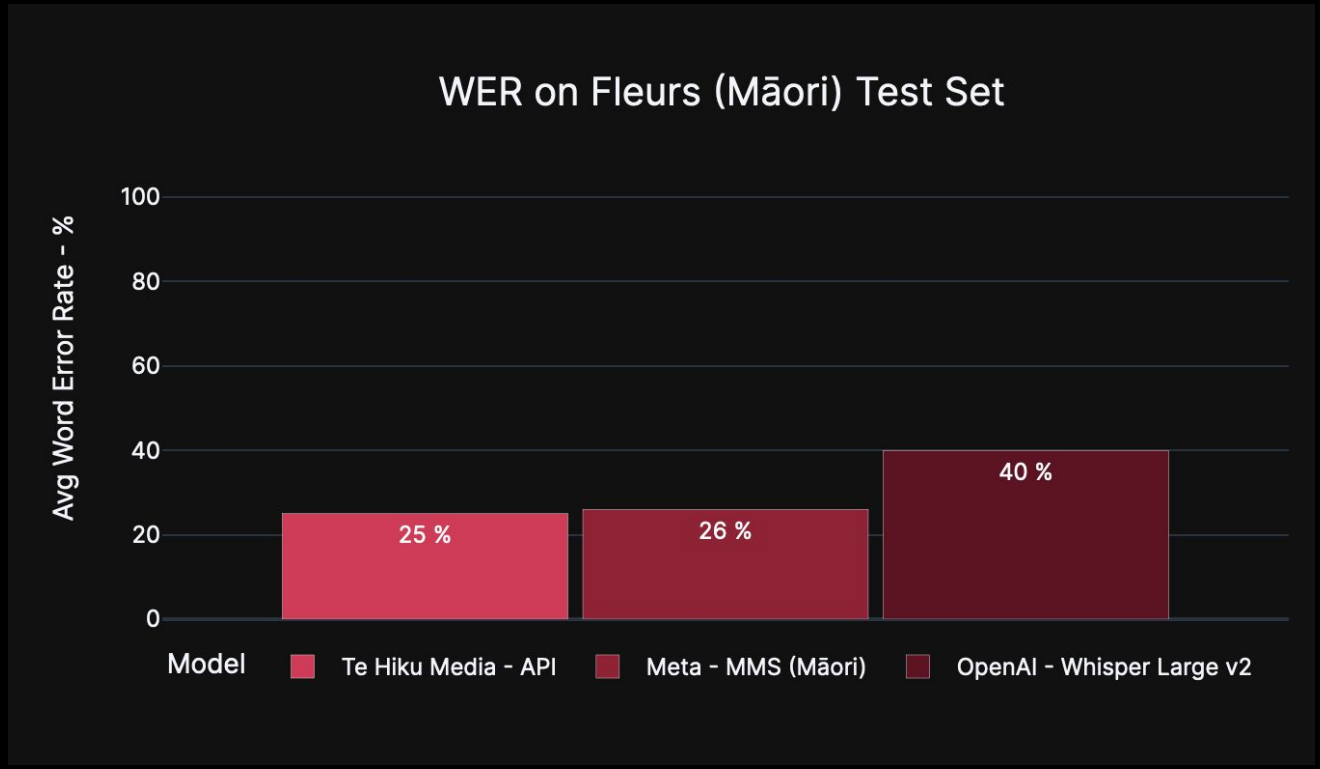
- **Training Data Quality**

  Training data includes Capitalization and Punctuation for enhanced language understanding, and readability

- **Iterations**

  Multiple iterations conducted to optimize performance and accuracy

How'd we do? Fleurs



Models

- Open AI Whisper
- Meta AI MMS
- Te Hiku Media API

Word error rate

- WER

Fleurs benchmark

- Bad pronunciation
- No code switching
- Limited Speakers
- Noisy audio

Fleurs Benchmark

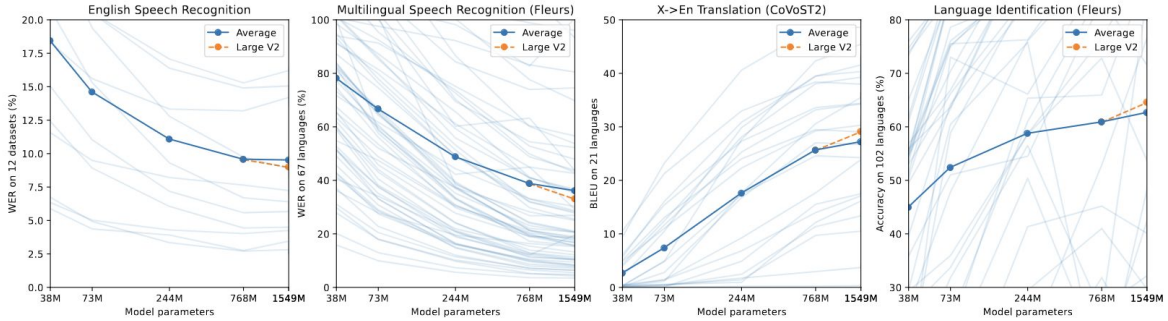


Figure 8. **Zero-shot Whisp**
lines represent individual da
Large V2 distinguished wi
analysis.

Model	Lingala	Lao	Lithuanian	Latvian	Maori	Macedonian	Malayalam	Mongolian	Marathi	Malay	Maltese	Myanmar	Norwegian	Nepali
Whisper tiny	105.4	115.1	98.5	91.6	94.5	73.3	101.5	113.7	100.3	51.2	100.8	124.8	62.0	101.8
Whisper base	96.7	105.1	87.3	79.8	77.5	59.9	107.4	125.7	100.3	35.1	97.6	122.6	44.0	102.4
Whisper small	91.3	102.2	65.6	53.2	59.5	36.9	100.9	144.2	60.2	18.9	92.2	110.1	24.2	69.5
Whisper medium	83.2	101.4	41.1	32.0	77.8	22.0	101.1	103.7	63.2	12.2	83.2	123.0	12.9	54.4
Whisper large	76.8	101.6	35.2	28.3	45.7	20.6	101.4	106.2	43.7	10.2	80.5	124.5	11.4	52.2
Whisper large-v2	75.6	101.5	28.1	23.1	38.5	16.5	100.7	110.5	38.3	8.7	76.6	115.7	9.5	47.1

Fleurs Benchmark



32057901684833978.wav

ki tērā atu taha he noa ngā āhuatanga tio hukapapa hoki ki ngā tini whenua ā ka haere tonu te ikiiki kāore he tino raru puta noa i te tau

Models

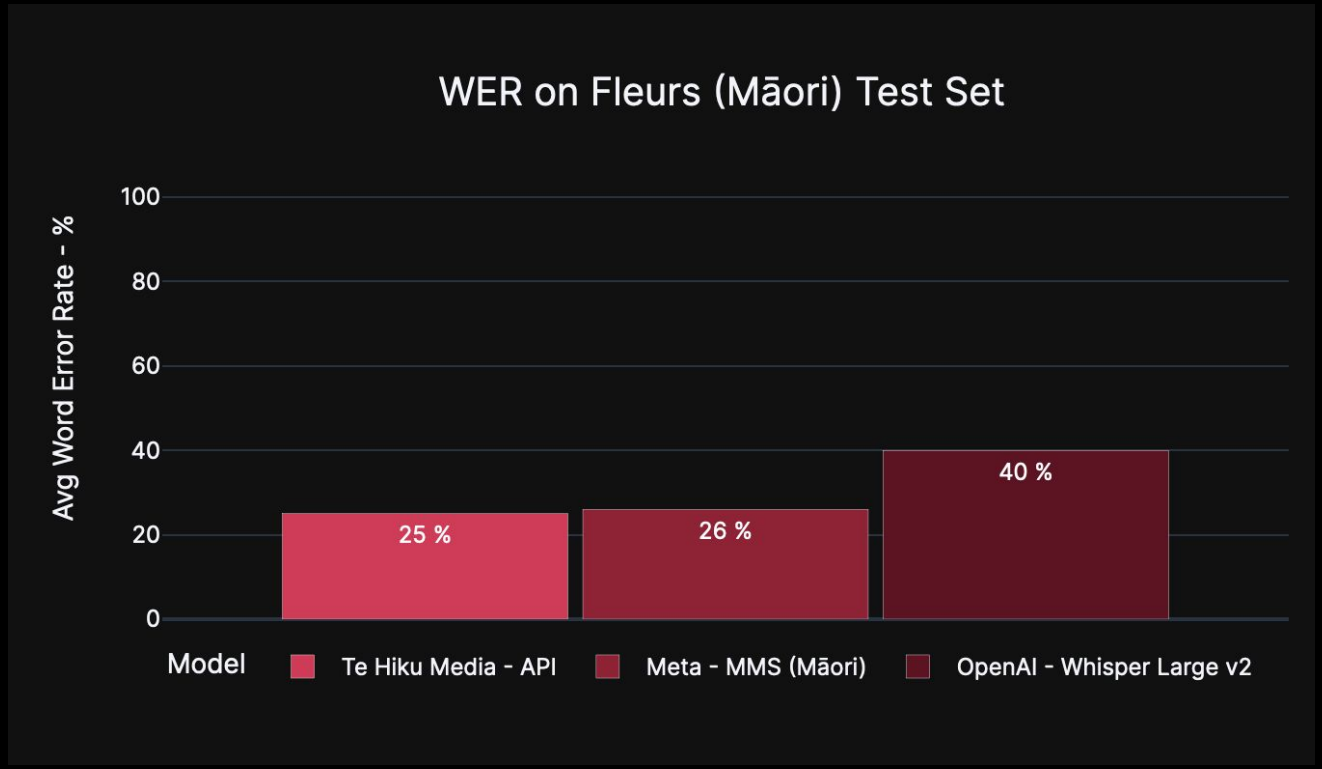
- Open AI Whisper
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Fleurs benchmark

- Bad pronunciation
- Limited Speakers
- Not native speakers
- Noisy audio
- Questionable data practices

- No code switching
- All lowercase
- No punctuation

How'd we do? Fleurs



Models

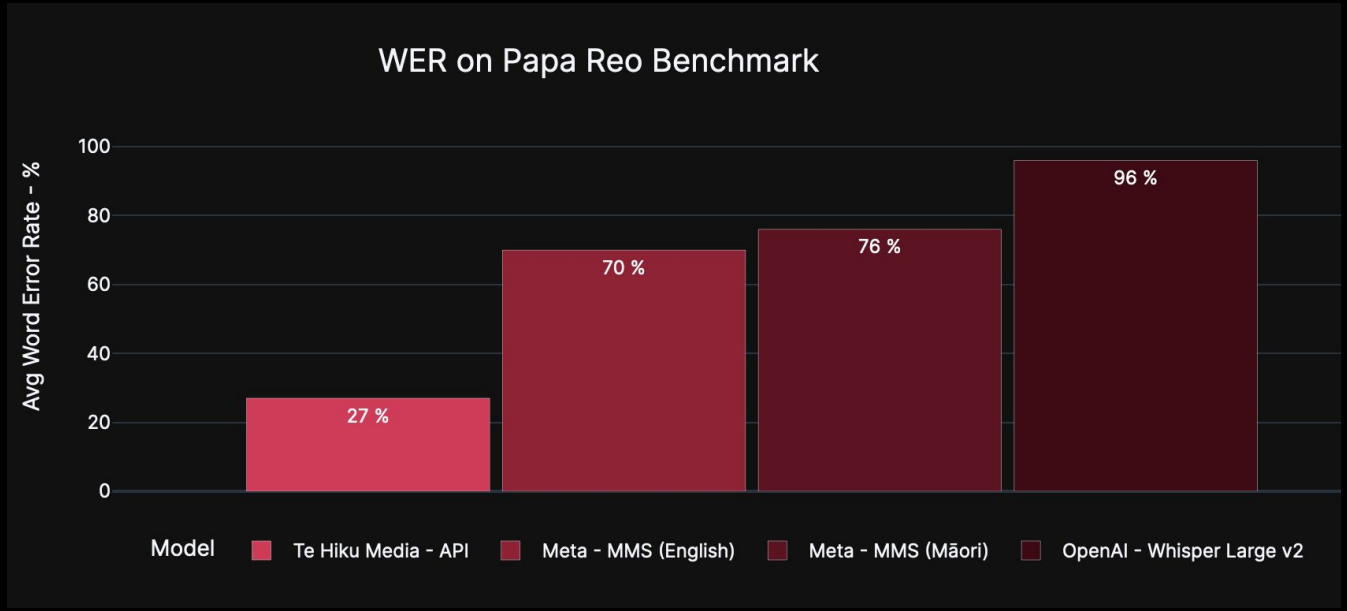
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Fleurs benchmark

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So how'd we do? Papa Reo



Models

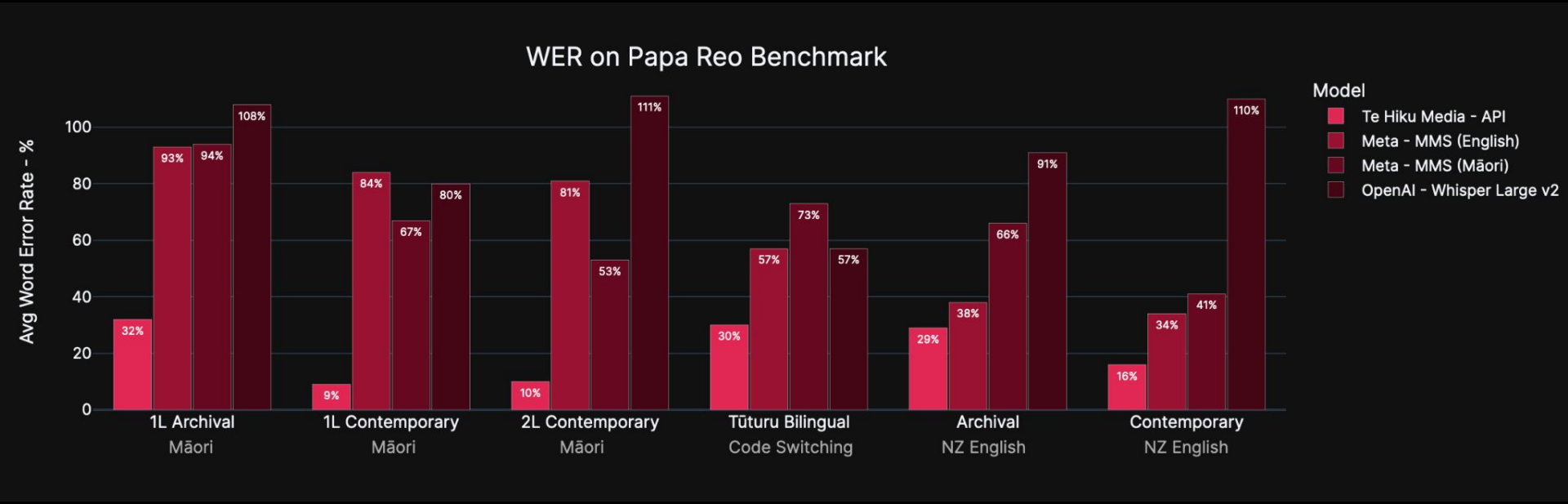
- Open AI Whisper
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Papa Reo Benchmark

- Harder
- Representative demographics
- Wide range of speakers
- Quality checked

- Bilingual
- Code switching
- Punctuation &c

So how'd we do? Papa Reo Benchmark



Tūturu Bilingual Code switching

Pā Henare Tate : segment_898_11

So in other words, it is the physical aspect, ka titiro mai anō koutou ki a au?

Pā Henare Tate : segment_898_12

Mōhio tonu atu koutou, nō Motutī ahau. So, you're beginning to look at my whole cultural background.

Contemporary NZ English:

With Loan Words & Proper Nouns

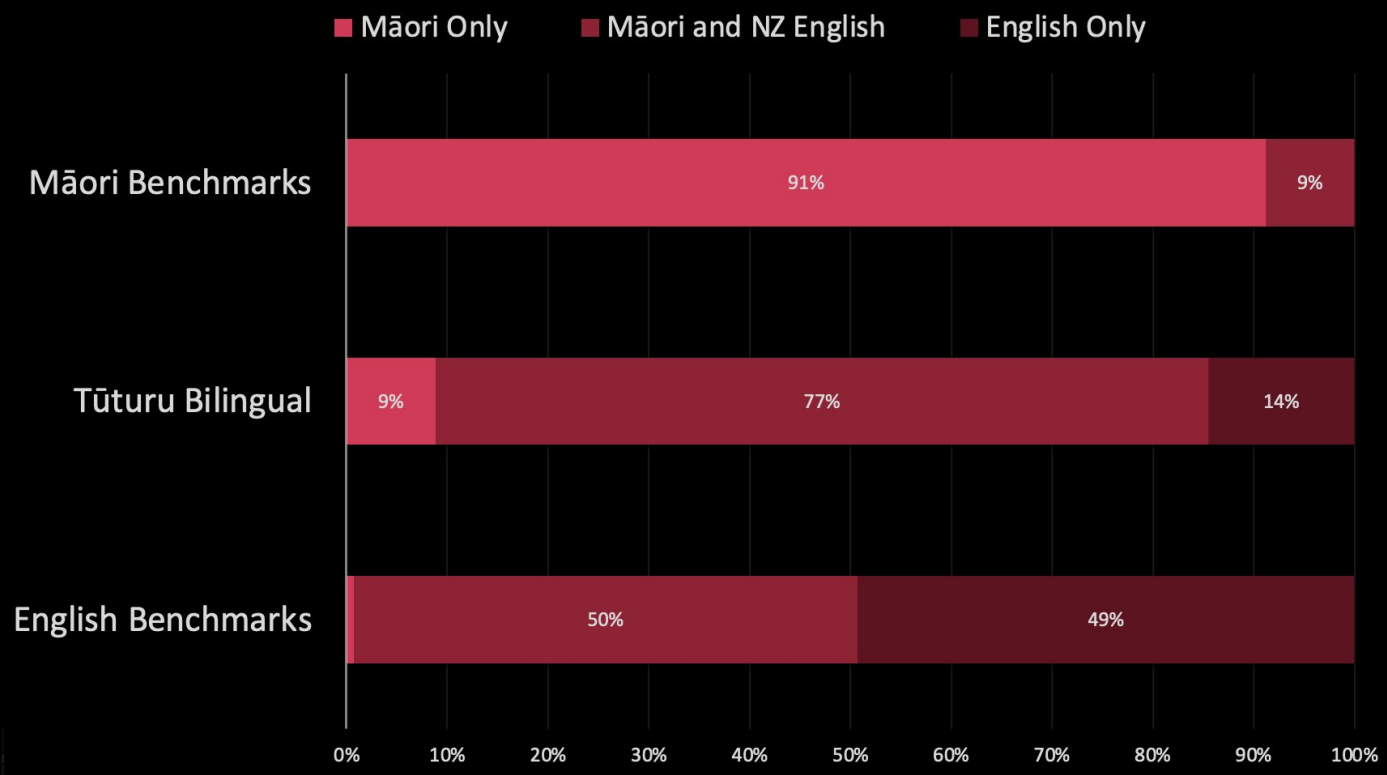
Jeol Bristow: segment_792_37

Absolutely and this is just some awesome news going into the weekend for a lot of our whānau as we have started back to work, back to school for 2022

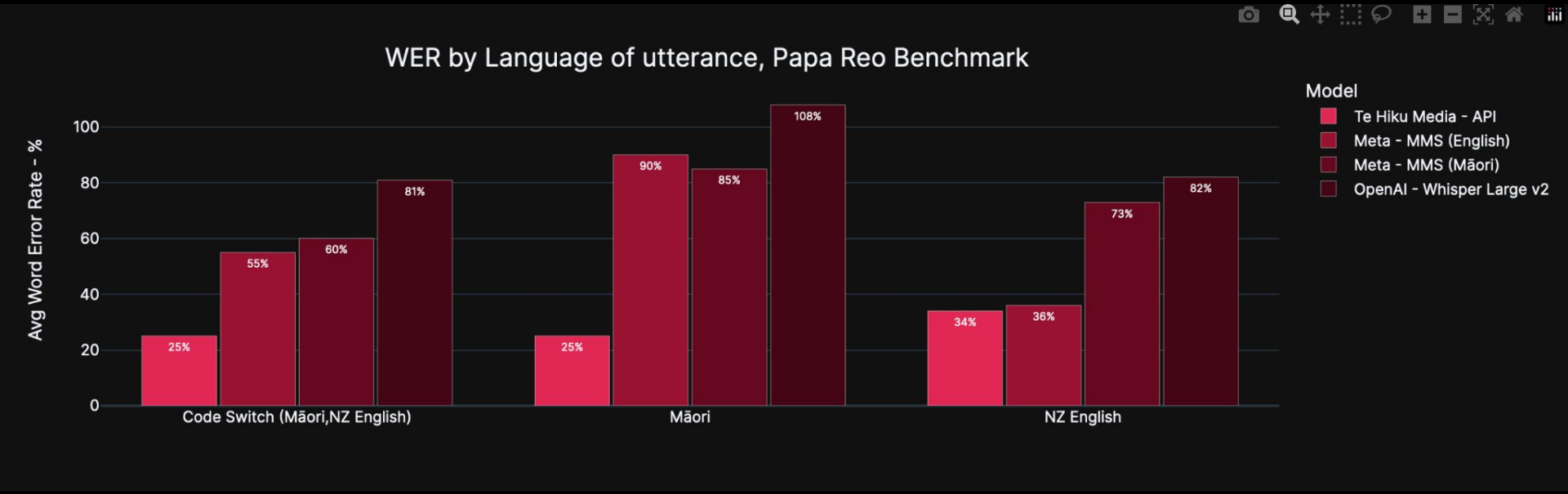
Jeol Bristow: radio segment_792_38

.. and to be able to have such a positive kaupapa just bestowed on all the Far North, Kaitāia.

Utterances containing words from one or more languages



Context Switching, Loan Words & Proper Nouns



Wrap up

How much data do you need?

- ~300 hrs+ is plenty to get going
- Build a basic model then use it to help you transcribe data and then train a better model (5000 hrs is better)

How much training and what architecture?

- You need six days on a GPU to train
- Conformer is a good model for relatively small amounts of data (even ~300 hrs)

How accurate ?

- If you concentrate on clean data and just your target languages **you** can get results that **meaningfully outperform** the latest models from the big companies on your own indigenous language

References

- **Baidu DeepSpeech2 (2016)**
  https://proceedings.mlr.press/v48/amodei16.pdf
- **Mozilla DeepSpeech (2017-2021)**
  https://github.com/mozilla/DeepSpeech
- **Open AI Whisper (Sep 2022)**
  https://openai.com/research/whisper
- **Papa Reo Conformer Bilingual API (Apr 2023)**
  https://papareo.io/docs
- **Meta AI MMS (May 2023)**
  https://research.facebook.com/publications/scaling-speech-technology-to-1000-languages

So how'd we do? Papa Reo kWER

